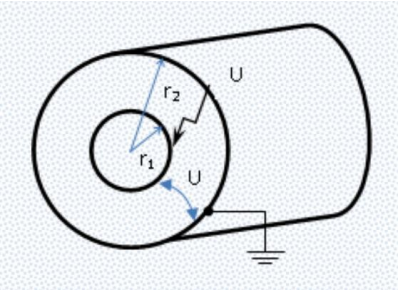
EE4047 – High Voltage Techniques I

New Topic:

Cylindrical Electrode Systems

Week 9.2 – Thursday Lesson



Asst. Prof. Dr. M. Alparslan

27.04.2023

OUTLINE

1 Introduction to Cylindrical Electrodes

2 Coaxial Cylindrical Electrode Systems

3 Multilayer Dielectric Coaxial Cylinders

4 Breakdown Behavior in Cylinderical Electrodes Systems

1 INTRODUCTION TO CYLINDRICAL ELECTRODES

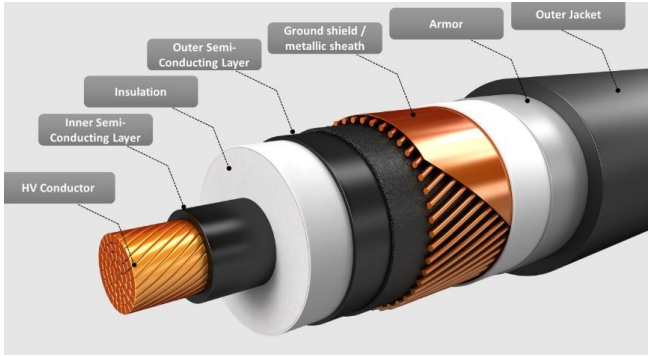
Two or more cylindrical electrodes with dielectric material between them form a cylindrical electrode system.

If they share the same axis, it is called coaxial, such as cables and bushings.

They may also be next to each other or crosswise like in transmission lines.

Moreover, many systems that have axial symmetry can be considered as a symmetrical electrode system. Some examples are motors, generators, transformers isolators, an overhead line and a ground.

The important parameters are electric field strength, electric flux density, potential and capacity.





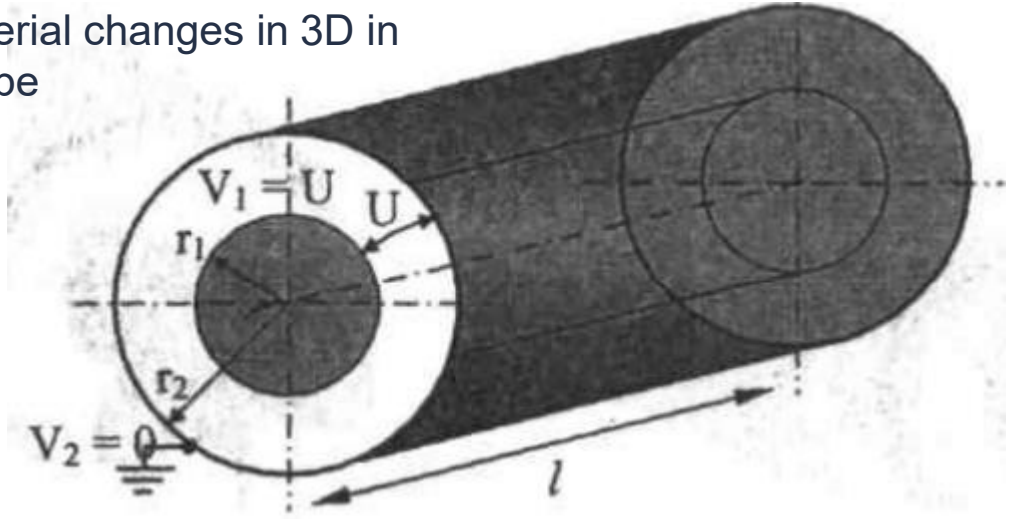
2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

Coaxial cylindrical electrode system forms the basis of cylindrical electrode systems.

Consider the system given in the figure with two electrodes with different radius values $r_1$and r_2 and the same length, l. The voltage applied to the inner electrode is U.

The potential on any point of the dielectric material changes in 3D in reality. However, due to axial symmetry, it can be analyzed in one dimension depending on r.

Considering a cylindrical coordinate system, a Laplace equation that depends only on r can be solved.



2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

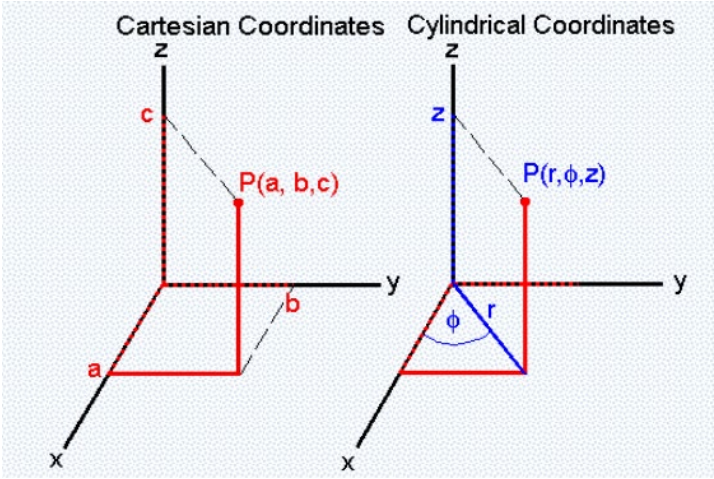
Considering a cylindrical coordinate system, a Laplace equation that depends only on r can be solved.

$$\Delta V = \frac{\partial^2 V}{\partial r^2} + \frac{1}{r}\frac{\partial V}{\partial r} + \frac{1}{r^2}\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

(dependent variable is r)

$$\Delta V = \frac{d^2 V}{dr^2} + \frac{1}{r}\frac{dV}{dr} = 0 \rightarrow V = V(r)$$

$$V = A + B.\ln(r)$$



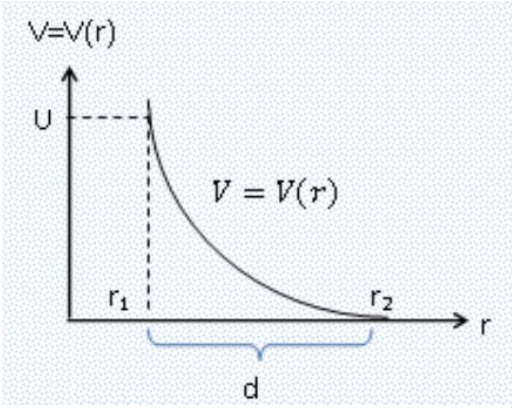
Giving the boundary conditions as $r = r_1 \quad V = U$ and $r = r_2 \quad V = 0$

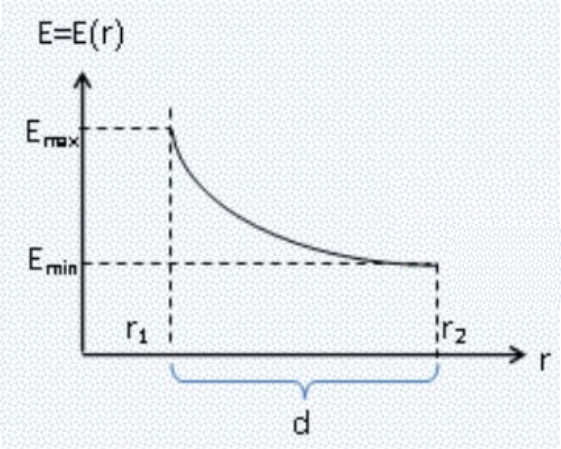
$V = \frac{U}{ln\frac{r_2}{r_1}} ln\frac{r_2}{r}$ with the relation $E = -grad\ V$ $\mathrm{E} = \frac{U}{ln\frac{r_2}{r_1}}\frac{1}{r}$

2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

$$V = \frac{U}{ln\frac{r_2}{r_1}} ln\frac{r_2}{r}$$

$$E = \frac{U}{ln\frac{r_2}{r_1}} \frac{1}{r}$$





For $r = r_1 \rightarrow$

$$E = E(r_1) = \frac{U}{r_1 \ln\frac{r_2}{r_1}} = E_{max}$$

For $r = r_2 \rightarrow$

$$E = E(r_2) = \frac{U}{r_2 \ln\frac{r_2}{r_1}} = E_{min}$$

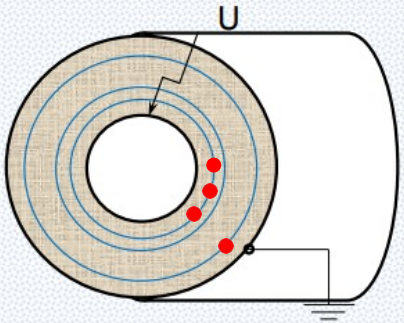
2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

$$E_{ave} = \frac{U}{ln\frac{r_2}{r_1}}\frac{1}{r_{av}}$$

$$r_{av} = \frac{r_2 - r_1}{ln\frac{r_2}{r_1}}$$

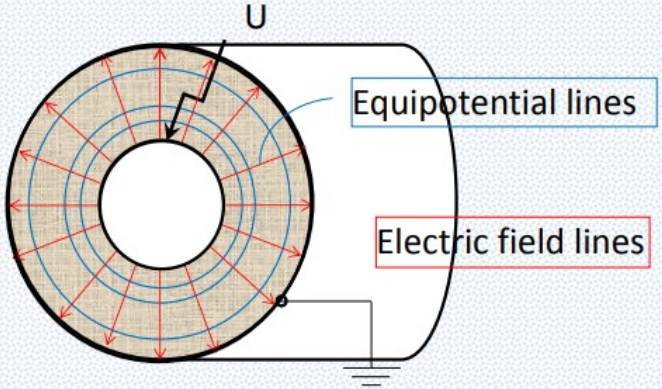
2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

Equipotential lines



Equipotential lines

Electric field lines



2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

The electric flux density can be calculated using the equation $D = Q\mathrm{T}S$

In the place of S, we can use the surface area formula of a cylinder: $S = 2\pi rl$

$$D = \frac{Q}{2\pi rl}$$

Using the relation $D = \varepsilon E$

$$E = \frac{Q}{2\pi\varepsilon l}\frac{1}{r}$$

Integral of the electric field gives us the voltage across the electrodes (U).

$$U = \int_{r_1}^{r_2} E\,dr = \frac{Q}{2\pi\varepsilon l}\int_{r_1}^{r_2}\frac{dr}{r} \quad \frac{Q}{2\pi\varepsilon l} \quad \frac{r_2}{r_1}$$

2 COAXIAL CYLINDRICAL ELECTRODE SYSTEMS

Capacity of a coaxial electrode system:

$$C = \frac{Q}{U} = \frac{2\pi\varepsilon l}{ln\frac{r_2}{r_1}} = \frac{2\pi\varepsilon_0\varepsilon_r l}{ln\frac{r_2}{r_1}}$$

In capacity calculations the unit for r_1 and r_2 is m, $\varepsilon_0 = 8{,}854.10^{-12}\ \mathrm{F/m}$, so the result (C) will be in Farad.

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Multilayer dielectric materials between the electrodes:

$$I = I_1 = I_2 = I_3 = \cdots = I_n$$

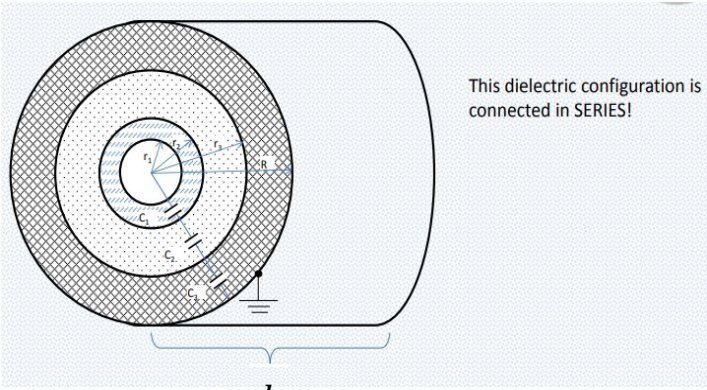
$$Q = Q_1 = Q_2 = Q_3 = \cdots = Q_n$$

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U = U_1 + U_2 + U_3 + \cdots + U_n$$

U is the voltage applied to the entire system

U_1, U_2, ..., U_n are voltages across the corresponding dielectric layers



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Voltages across each dielectric layer:

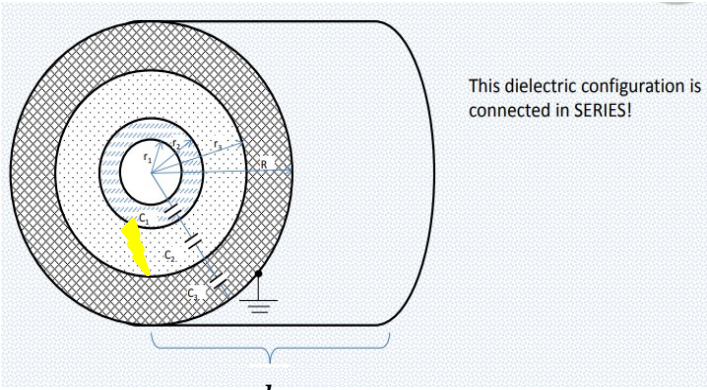
$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U_1 = \frac{C\,U}{C_1}$$

$$U_2 = \frac{C\,U}{C_2}$$

$$\vdots$$

$$U_n = \frac{C\,U}{C_n}$$



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

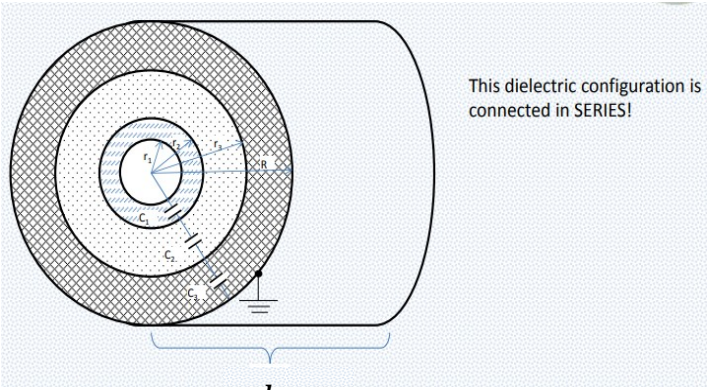
Equivalent capacitance of the system:

$$X_C = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

$$\frac{1}{\omega C} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2} + \frac{1}{\omega C_3} + \cdots + \frac{1}{\omega C_n}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$

C : equivalent capacitance of the system, total capacitance of the system
U : voltage applied to the entire system
C_i : capacitance of the ith dielectric layer
U_i : voltage across i^{th} dielectric layer



END OF THE LESSON

Any questions?

Check out the following two electric field videos on Youtube:

https://www.youtube.com/watch?v=re6iFu5SIxY

https://www.youtube.com/watch?v=mdulzEfQXDE

Check out our playlists on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ